DOCKHUB · TRACEABILITY MATRIX

DockHub traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Generated by .kit/trace.py

DockHub traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 13 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Fast swap	DKH-CAL-001, DKH-BLD-001	Task analysis; later timed trials with riders	Met (31 s)
UR-005	PR-002	SwapCell compatibility	DKH-CAL-001, DKH-BLD-001	Interface review with the SwapCell project	Met by design review; one CAN channel per bay
UR-002	PR-003	Charge speed	DKH-CAL-001, DKH-BLD-001	Charge profile calculation	Met (1.2 h and 2.3 h)
UR-002	PR-004	Throughput	DKH-CAL-001	Day model with a demand profile	Met by the four-bay fit (20 of 20 served, longest wait 4 min; ceiling 36 a day). The two-bay first prototype serves 9 of 20
UR-009	PR-005	Grid supply	DKH-CAL-001, DKH-BLD-001, DKH-STM-001, BOM	Load calculation	Met (1.25 kW; 11.0 A at 120 V) with chargers of power factor 0.9 or better Independent review (DKH-STM-001): at 10 % low mains voltage constant-power chargers draw 12.2 A at 120 V, over the 12 A continuous limit if all four run; the simulated day runs at most 3 (8.3 A); low-voltage case to add to DKH-CAL-001
UR-013	PR-006	Solar contribution	DKH-CAL-001, DKH-STM-001	Day model	Met on paper, thin margin (9.3 %; 11.4 % if flat packs are routed to bay 1 in daylight, which turns one rider in 20 away, so normal routing is kept; 16.1 % if the solar output is later switched to the flattest pack, a TRL 4 trial) Basis: 9.3 % is measured at the pack terminals; at the inputs (0.77 kWh panel against 8.04 kWh grid to chargers) it is 8.7 %, on one clear design day; the definition is proposed, awaiting Amish (DKH-DEC-001 O6)
UR-003	PR-007	Fire detection and response	DKH-CAL-001, DKH-BLD-001, DKH-STM-001	Latency budget; later bench test with a heat source	Not verifiable at TRL 3 (electronic latency 3.1 s, contactor 0.15 s)
UR-003	PR-008	Fire containment	DKH-CAL-001, DKH-STM-001,	Later propagation	At risk**; not verifiable at TRL 3

User requirement	Product requirement	Title	Evidence	Verification	Status
			DKH-OPT-001	test by an accredited lab	
UR-004	PR-009	Climate	DKH-CAL-001, DKH-BLD-001, DKH-STM-001, DKH-OPT-001	Thermal calculation; later chamber test	At risk**: uninterrupted charging needs ambient below about 29 to 39 °C (pack charge limit 45 °C), so the warmest bay may pause just under 30 °C; cold-soaked packs cannot start below about -1 °C, as now stated IP54 is not demonstrated: the vent slots and filtered intake make it doubtful until an enclosure test (STEM S27); the lowest calculated limit for uninterrupted charging is 29.4 C (S9)
UR-007	PR-010	Security	DKH-CAL-001, DKH-BLD-001	Design review	Met by design review
UR-008	PR-011	Accessibility and use	DKH-CAL-001, DKH-BLD-001, DKH-STM-001	Model check; later user trials	Met (0.73 to 1.10 m)
UR-010	PR-012	Privacy and data	DKH-CAL-001	Design review	Met by design review (token list of 200 kB, including anonymous prepaid cards with a pack deposit)
UR-009	PR-013	Sidewalk footprint	DKH-CAL-001, DKH-BLD-001, DKH-STM-001	Model check	Met (1.0 x 0.5 m with a flush plinth; lowest canopy point over the sidewalk 2.17 m)
UR-011	PR-014	Maintainability	DKH-CAL-001, DKH-BLD-001	Design review; later timed trial	Not verifiable at TRL 3
UR-006	PR-015	Wind and anchoring	DKH-CAL-001, DKH-STM-001	Wind load calculation	Met on paper: four M12 stainless wedge anchors (3.8 kN design tension each) into a 1,400 x 1,100 x 300 mm cast pad of about 1,109 kg; equilibrium factor 1.12 at a 30 m/s gust. The anchor rating is confirmed from the maker's data when bought Upright only; STEM S16 is plausible (6 % margin; anchor design tension 4.13 kN with matching factors, proposed 4.2 kN, DKH-DEC-001 O5). The canopy being intact (panel clamps,

User requirement	Product requirement	Title	Evidence	Verification	Status
					roof pull-through at 1,251 N per post) and wind from other directions are not yet checked
UR-012	PR-016	Cost	DKH-CAL-001	Priced BOM (`bom/bom.csv`)	Over the value-engineering target by $319** ($1,519 against $1,200: $1,214 for the buildable cabinet, DKH-DDR-003, plus $305 for the anchors and site pad chosen on 2026-10-02; $1,743 with all four bays fitted)
UR-003	PR-017	Aerosol is kept in	DKH-STM-001	Design review; retention test in a bay mock-up (TRL 4)	Met by design (control rule); retention time to measure at TRL 4 (S13); gas build-up with the fans stopped is not yet assessed (S29, DKH-DEC-001 O4, proposed, awaiting Amish)

Gaps

None.